A Brief Review of Quantum Liouville Theory

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Outline

- The Quantum Liouville Theory
- Degenerate Fields & Operator Algebra
- Logarithmic Conformal Field Theory
- Comparison with fermionic theory
- Conclusion & Future Work

The Quantum Liouville Theory

The Quantum Liouville Theory

- Setup: Partition function of 2D quantum gravity coupled to conformal matter

$$Z = \int D[g] e^{-S_{matter}[g]}$$

- Conformal gauge fixing using background metric

$$\hat{g}^{\mu\nu}(x) = e^{\hat{\sigma}(x)} g^{(0)}_{\mu\nu}(x)$$

- Problem and solution

$$\langle T^{\mu}_{\mu} \rangle \neq 0$$

Sum over the scale factor σ $\Longrightarrow$ Functional integral is not standard "linear" measure as well as trace anomaly!

Perform the Jacobian and found it can be exponentiated into a local action (by Polyakov (1981))

$$D[g] = \underbrace{J[\sigma]}_{\text{anomaly}} D[\sigma] = e^{S_L[\sigma]} D[\sigma]$$

The Quantum Liouville Theory

- The Classical Liouville Theory

$$S_{\mathrm{cl}} = \frac{1}{4\pi}\int d^2\,x\sqrt{g}\left((\partial\varphi)^2 + \Lambda e^{\varphi}\right)$$

- Equation of Motion

$$\partial\bar{\partial}\varphi = \frac{\Lambda}{2}e^{\varphi} \Longrightarrow$$ surfaces of **constant negative curvature**

Solvable by Bäcklund transformation!

The Quantum Liouville Theory

- The renormalized action in Liouville Theory

$$A_L[\hat{g},\varphi] = \frac{1}{4\pi}\int \sqrt{\hat{g}}\left[\hat{g}^{\mu\nu}\partial_\mu\varphi\partial_\nu\varphi + \overbrace{Q\hat{R}\varphi}^{\text{background charge}} + \underbrace{4\pi\mu e^{2b\varphi}}_{\text{Liouville Interaction}}\right]d^2x$$

kinetic

- Conformal gauge fixing using background metric

$$\hat{g}^{\mu\nu}(x) = e^{\hat{\sigma}(x)} g^{(0)}_{\mu\nu}(x) \quad \text{with geometric definition } \sigma(x) = 2b\varphi$$

- The functional integral (Matter + Ghosts + Liouville)

$$Z_{matter,ghosts}[\hat{g}]e^{-A_L[\hat{g},\varphi]}D_{\hat{g}}[\varphi]$$

The Quantum Liouville Theory

- Requirement: Background independence of metric
- Constraint 1: Central charge balance --- **Conformal Invariant** with a total central charge of zero

$$c_{ghost}(-26) + c_{matter} + c_{Liouville} = 0 \quad \Longrightarrow \quad Q = \sqrt{\frac{25 - c_M}{6}}$$

- Constraint 2: Marginality --- Interaction term be conformal invariant and dimensionless ($\Delta = 1$)

$$\Longrightarrow \quad Q = \frac{1}{b} + b$$

The Quantum Liouville Theory

- The holomorphic and anti-holomorphic energy-momentum tensor with respect to the background metric

$$T = -(\partial_z \varphi)^2 + Q\partial_z^2 \varphi$$
$$\bar{T} = -(\partial_{\bar{z}} \varphi)^2 + Q\partial_{\bar{z}}^2 \varphi$$

- Operator Product Expansion

$$T(z)T(w) = \frac{c_L/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \quad \text{with} \quad c_L = 1 + 6Q^2$$

- Duality

$$Q = b + \frac{1}{b} \quad \Longrightarrow \quad \text{Invariant for} \quad b \leftrightarrow \frac{1}{b}$$

The Quantum Liouville Theory

- Primary field corresponding to momentum α in Liouville Theory

$$V_\alpha(\mathrm{x}) = e^{2\alpha\varphi(x)}$$

- Conformal dimension using OPE

$$T(z)V_\alpha(w,\bar{w}) \sim \frac{\alpha(Q-\alpha)}{(z-w)^2}V_\alpha(w,\bar{w}) + \frac{1}{z-w}\partial_w V_\alpha(w,\bar{w}) \Longrightarrow \Delta_\alpha = \alpha(Q-\alpha)$$

Symmetry here!

$$\Delta_\alpha = \Delta_{Q-\alpha}$$

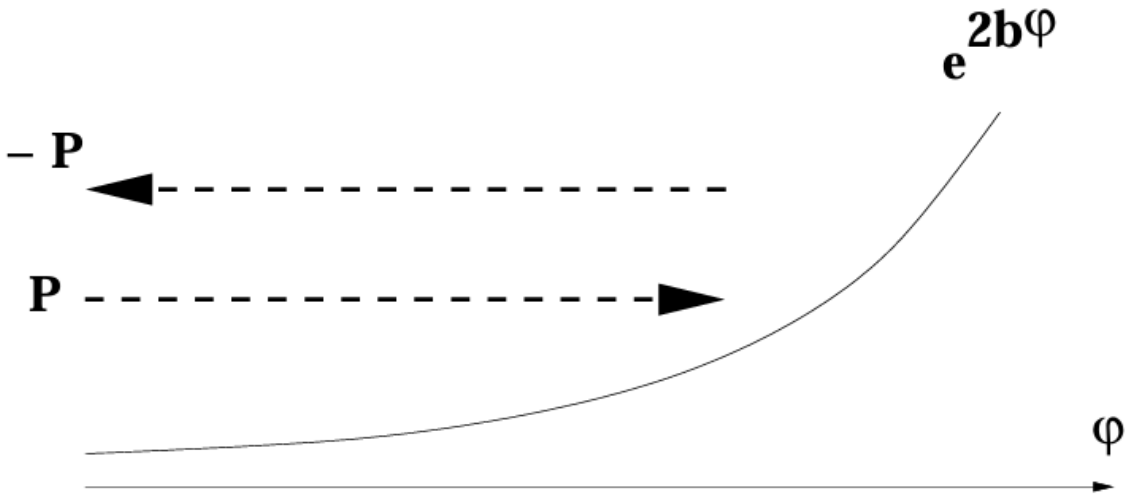
- Seiberg bound: physical spectrum is continuous with $\alpha = \frac{Q}{2} + iP$

- $\mathrm{Re}(\alpha) \leq Q/2$: Normalizable states
- $\mathrm{Re}(\alpha) > Q/2$: Non-normalizable (Local singularity)

The Quantum Liouville Theory

- Reflection property: $\Delta_\alpha = \Delta_{Q-\alpha}$

  $V_\alpha = S(\alpha)V_{Q-\alpha}$ (here $S(\alpha)$ has analytic form)
- Interpretation: The exponential potential acts like a "wall" at $\phi \to \infty$



The Quantum Liouville Theory

- Three-point function in Liouville Theory: DOZZ Formula (1990)

$$\langle V_{\alpha_1}(x_1)V_{\alpha_2}(x_2)V_{\alpha_3}(x_3)\rangle = \frac{C(\alpha_1,\alpha_2,\alpha_3)}{(x_{12}\bar{x}_{12})^{\Delta_1+\Delta_2-\Delta_3}(x_{23}\bar{x}_{23})^{\Delta_3+\Delta_2-\Delta_1}(x_{31}\bar{x}_{31})^{\Delta_3+\Delta_1-\Delta_2}}$$

with $C(\alpha_i) \propto \dfrac{\Upsilon'(0)\prod\Upsilon(2\alpha_i)}{\Upsilon(\sum\alpha_i - Q)\prod(\alpha_i+\alpha_j-\alpha_k)}$

have explicit form

Degenerate Fields & Operator Algebra

Degenerate Fields & Operator Algebra

- Classical Liouville Theory: Equation of motion

$$\partial\bar{\partial}\varphi = Me^{\varphi} \qquad \text{Non-linear!}$$

- Linear equation? Basic example of n=2 case:

$$\left(\partial^2 + T^{(c)}\right)e^{-\varphi/2} = 0$$

which satisfies second-order linear equation, manifesting of a "degenerate" field

- General construction of degenerate field

$$V_n^{(c)} = e^{(1-n)\varphi/2} \; with \;\; n = 1,2,3 \ldots \quad \Longrightarrow \quad D_n^{(c)} V_n^{(c)} = \overline{D}_n^{(c)} V_n^{(c)} = 0$$

null vector

with a unique differential operator

$$D_n^{(c)} = \partial^n + \cdots + (n-1)\partial^{n-2}T^{(c)}$$

Degenerate Fields & Operator Algebra

- Next, consider the following field $\varphi V_n^{(c)}$
- Property of this new field

$$\overline{D}_n^{(c)} D_n^{(c)}\left(\varphi e^{(1-n)\varphi/2}\right) = B_n^{(c)} e^{(1+n)\varphi/2} \longrightarrow \text{classical higher equation of motion}$$

with a particular coefficient

$$B_n^{(c)} = 2(-1)^{n+1} n!\,(n-1)!\left(\frac{M}{2}\right)^n$$

- The new field derives from the derivative with respect to n!
- The equation maps a “Logarithmic field” to a “dual” exponential field

Degenerate Fields & Operator Algebra

- Next, consider the quantum Liouville field theory Lagrangian density

$$L = \frac{1}{4\pi}(\partial_a \phi)^2 + \mu e^{2b\phi} \Longrightarrow \partial\bar{\partial}\phi = \pi b \mu e^{2b\phi}$$

- Parameter mapping

$$2b\phi \to \varphi, \qquad T \to b^{-2}T^{(c)}, \qquad 2\pi\mu b^2 \to M$$

- Discrete subset of primaries $V_{m,n} = V_{\alpha_{m,n}}$ with special value of positive m and n

$$\alpha_{m,n} = -(m-1)\frac{1}{2b} - (n-1)\frac{b}{2}$$

these fields called degenerate, since the corresponding conformal dimension

$$\Delta_{m,n} = \frac{Q^2}{4} - \frac{(mb^{-1} + nb)^2}{4}$$

are the Kat dimensions of the degenerate representations of the Virasoro algebra

Duality!

$$b \leftrightarrow \frac{1}{b}$$

$$\Downarrow$$

$$\alpha_{m,n}(b) \leftrightarrow \alpha_{n,m}(1/b)$$

Degenerate Fields & Operator Algebra

- Null vector property: $V_{m,n}$ has a null vector as level mn

$$D_{m,n}V_{m,n} = \overline{D}_{m,n}V_{m,n} = 0$$

with the operator constructed by

$$D_{m,n} = L_{-1}^{mn} + d_1^{m,n}(b)L_{-2}L_{-1}^{mn-2} + \cdots$$

- Example of Level 2 (1,2)

$$D_{1,2} = L_{-2} + b^2 L_{-1}^2 \quad \Longleftrightarrow \quad \partial^2 + T^{(c)} \text{ in classical}$$

Logarithmic Conformal Field Theory

Logarithmic Conformal Field Theory

- What is Logarithmic Conformal Field Theory?

Standard CFT:
Diagonalizable Hamiltonian (L_0)

$$L_0|\Delta\rangle = \Delta|\Delta\rangle$$

primary field

$\Longrightarrow$

Logarithmic CFT:
Non-diagonalizable Hamiltonian (Jordan Block)

$$L_0|\phi\rangle = \Delta|\phi\rangle \quad \text{primary field}$$

$$L_0|\psi\rangle = \Delta|\psi\rangle + |\phi\rangle$$

logarithmic partner

$$\begin{bmatrix} \Delta & 1 \\ 0 & \Delta \end{bmatrix}$$

Logarithmic Conformal Field Theory

- Signature of Logarithmic CFT: correlation function (Scale invariance is broken by logarithmic terms)

$\langle \phi(z)\phi(0)\rangle = 0$ ⟶ zero norm for primaries

$$\langle \phi(z)\psi(0)\rangle = \frac{B}{z^{2\Delta}}$$

logarithmic coupling (zero in standard CFT)

$$\langle \psi(z)\psi(0)\rangle = \frac{C - 2B\ln z}{z^{2\Delta}}$$

Liouville Theory as Logarithmic Conformal Field Theory

- What is the logarithmic partner of V_α in Liouville Theory?

$$\boxed{V'_\alpha = \frac{1}{2}\frac{\partial}{\partial\alpha}V_\alpha = \phi e^{2\alpha\phi}} \Longrightarrow \text{logarithmic field (by Al.Zamolodchikov)}$$

- OPE of this logarithmic field

$$T(z)V'_\alpha(w,\bar{w}) = \frac{\alpha(Q-\alpha)}{(z-w)^2}V'_\alpha(w,\bar{w}) + \frac{(Q/2-\alpha)}{(z-w)^2}V_\alpha(w,\bar{w}) + \frac{1}{z-w}\partial_w V'_\alpha(w,\bar{w})$$

Jordan block here!

Liouville Theory as Logarithmic Conformal Field Theory

- Property of $V'_{m,n}$

$$D_{m,n}V_{m,n} = 0 \xrightarrow{\;?\;} D_{m,n}V'_{m,n} = 0$$

- The answer is no! It is a primary field with different conformal dimension!

Proof: $D_{m,n}\overline{D}_{m,n}V_{m,n} = 0$ is a null vector

Consider the vicinity of $\alpha = \alpha_{m,n}$ $\longrightarrow$ $D_{m,n}\overline{D}_{m,n}V_\alpha = 0 + (\alpha - \alpha_{m,n})A_{m,n} + \cdots$

$$A_{m,n} = \lim_{\alpha\to\alpha_{m,n}} \frac{D_{m,n}\overline{D}_{m,n}V_{m,n}}{\alpha - \alpha_{m,n}}$$

is also a primary field

Apply the derivative $\longrightarrow$ $D_{m,n}\overline{D}_{m,n}V'_{m,n} = A_{m,n}$

Primary field with conformal dimension $\Delta_{m,n} + mn$!

$$D_{m,n} = L_{-1}^{mn} + d_1^{m,n}(b)L_{-2}L_{-1}^{mn-2} + \cdots$$

Liouville Theory as Logarithmic Conformal Field Theory

- What is the new primary field after mapping by $D_{m,n}\overline{D}_{m,n}$?

$$D_{m,n}\overline{D}_{m,n}V'_{m,n} = B_{m,n}\tilde{V}_{m,n} \longrightarrow \text{Denote } \tilde{V}_{m,n} = V_{\tilde{\alpha}_{m,n}} \text{ with } \tilde{\alpha}_{m,n} = -(m-1)\frac{1}{2b} + (n+1)\frac{b}{2}$$

Operator-valued relation

having conformal dimension $\Delta_{m,n} + mn$ as we expect!

$\downarrow$

$\tilde{V}_{m,n} = V_{m,-n}$

Dual field by flipping the sigh of n

Analytic solution: $$B_{m,n} = \left(\pi\mu\gamma(b^2)\right)^n b^{1+2n-2m}\gamma(m-nb^2) \prod_{\substack{k=1-n \\ l=1-m \\ (k,l)\neq(0,0)}}^{\substack{m-1 \\ n-1}} (lb^{-1}+kb)$$

Comparison with fermionic theory

Comparison with fermionic theory

- Bosonization between non-Hermion system

Dirac fermion with non-Hermiton mass-term ($c = -2$) $\boxed{L = \text{Dirac Term} + \Delta\psi_L^\dagger\psi_R}$

$\Downarrow$ Bosonization
(project to fermion parity even states)

Non-Hermion Bosonic Field $\boxed{L = \frac{1}{4\pi}(\partial_a\phi)^2 + \Delta e^{2b\phi}}$ with $b = \frac{i}{\sqrt{2}} \Longrightarrow c = -2$

Liouville Theory!

Comparison with fermionic theory

- Indecomposability parameter --- Characterizes the structure of the Jordan cell

Symplectic Fermion CFT

Classical Liouville Theory

Level-1: $\beta_1 = -1$ →

Level-2: $\beta_2 = -18$ →

Matched!

$$\bar{D}_1^{(c)} D_1^{(c)} \varphi = M e^{\varphi}$$

$$\bar{D}_2^{(c)} D_2^{(c)} \left(\varphi e^{-\varphi/2}\right) = -M^2 e^{3\varphi/2}$$

$$\bar{D}_3^{(c)} D_3^{(c)} \left(\varphi e^{-\varphi}\right) = 3M^3 e^{2\varphi}$$

$$\bar{D}_4^{(c)} D_4^{(c)} \left(\varphi e^{-3\varphi/2}\right) = -18M^4 e^{5\varphi/2}$$

$$\bar{D}_5^{(c)} D_5^{(c)} \left(\varphi e^{-2\varphi}\right) = 180M^5 e^{3\varphi}$$

Comparison with fermionic theory

- Staggered module

Fermionic case

$$\begin{tikzcd}
 & \left|\psi_L^1\right\rangle \arrow[dl, "\beta_1^{-1}A_1^\dagger"'] \arrow[dd, "L_0 - I"] \\
\left|\xi_L^1\right\rangle \arrow[dr, "A_1"'] & \\
 & \left|\phi_L^1\right\rangle
\end{tikzcd}$$

logarithmic coupling

$\beta_1 = -1$

$\Longrightarrow$

Bosonic case

$$\begin{tikzcd}
 & V'_{m,n} \arrow[dl, "B_{m,n}^{-1}D_{m,n}\bar{D}_{m,n}"'] \arrow[dd, "L_0 - \Delta_{m,n}I"] \\
\tilde{V}_{m,n} \arrow[dr, "?"'] & \\
 & V_{m,n}
\end{tikzcd}$$

logarithmic coupling

$\left(D_{m,n}\bar{D}_{m,n}\right)^\dagger \ldots ?$

Comparison with fermionic theory

Bosonic case

$$\begin{tikzcd}
 & V'_{m,n} \arrow[dl, "B^{-1}_{m,n} D_{m,n} \bar{D}_{m,n}"'] \arrow[dd, "L_0 - \Delta_{m,n} I"] \\
\tilde{V}_{m,n} \arrow[dr, "?"'] & \\
 & V_{m,n}
\end{tikzcd}$$

$(D_{m,n}\bar{D}_{m,n})^{\dagger} \ldots ?$

$$L_n = L^{\dagger}_{-n} \longrightarrow (D_{m,n}\bar{D}_{m,n})^{\dagger}\tilde{V}_{m,n} = 0 \nearrow \text{null vector}$$

since dual fields are primary fields

$$\boxed{\text{Only Property right now: } \tilde{V}_{m,n} = V_{m,-n}}$$

Conclusion & Future Work

Conclusion & Future Work

- Conclusion

1. Logarithmic Liouville theory is the bosonization of symplectic fermions with non-Hermitian structure.
2. Liouville field theory has Jordan structure between primary fields and logarithmic fields
3. Operator-valued relation establish a dual primary field in Liouville field theory

- Future Work

1. Establish the zero mode in Liouville field theory and understand its Hilbert space
2. Form a staggered module and calculate the indecomposability parameter
3. Interpret the Kat indices m and n in terms of fermionic twist field

Reference

- Al. Zamolodchikov, "Higher Equations of Motion in Liouville Field Theory," in *Proceedings of the VI International Conference 'CFT and Integrable Models'* (Chernogolovka, Russia, 2002); arXiv:hep-th/0312279
- A. Zamolodchikov and Al. Zamolodchikov, *Lectures on Liouville Theory and Matrix Models*
- V. Gurarie, "Logarithmic Operators in Conformal Field Theory," *Nucl. Phys. B* **410**, 535-549 (1993); arXiv:hep-th/9303160.